The Grandparent Care Economy: How Co-Resident Grandparents Reduce Women's Childcare Burden in Urban India

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Abstract

Using India's 2024 Time Use Survey, we examine whether co-resident grandparents reduce the childcare burden on mothers in urban households. We find that in urban multi-generational households, women spend substantially less time per day on childcare compared to similar households without grandparents—even after accounting for any eldercare burden they may provide to those grandparents. This "grandparent dividend" is particularly pronounced for working mothers, who experience larger reductions in childcare time than non-working mothers. Grandmothers provide more childcare support than grandfathers. These findings highlight the economic value of intergenerational co-residence in India's rapidly urbanizing context, where formal childcare infrastructure remains scarce. However, the grandparent dividend disappears in rural areas, where joint family structures serve different economic functions. Our results have implications for understanding female labor force participation, aging policy, and the changing economics of multi-generational households.

JEL Classification: J12, J13, J14, J16, D13

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Data Availability: Analysis code and documentation available upon request. Original data from National Statistical Office Time Use Survey 2024.

1 Introduction

India's female labor force participation rate—at 24%—is among the lowest in the world, lower even than many Middle Eastern countries (World Bank 2023). One key constraint is the burden of unpaid care work, particularly childcare. Indian women spend an average of 5.5 hours per day on unpaid domestic and care work, compared to just 1.5 hours for men (NSSO 2020). Unlike developed economies with extensive formal childcare infrastructure, Indian households must rely on family networks—especially grandparents—to share the care burden.

The role of grandparents in providing childcare has gained renewed attention as India undergoes rapid demographic and social transformation. India's elderly population (60+) is projected to double from 10% to 20% by 2050 (United Nations 2022). Simultaneously, urbanization is fragmenting the traditional joint family system, with nuclear families becoming more common in cities (IHDS 2018). Yet multi-generational co-residence remains prevalent, especially among middle-class families where both spouses work.

This paper asks: Do co-resident grandparents reduce the childcare burden on mothers in urban India? And if so, is this "grandparent dividend" offset by increased eldercare responsibilities?

Using India's 2024 Time Use Survey, we compare urban households with and without co-resident grandparents (defined as parents or parents-in-law of the household head). We focus on urban areas because: (1) urban women face higher opportunity costs of time spent on childcare, given better labor market opportunities; (2) urban areas lack the extended kin networks available in villages; (3) formal childcare remains prohibitively expensive for most urban households (Desai and Joshi 2019).

We find strong evidence of a **grandparent childcare dividend** in urban India:

1. **Women in urban households with grandparents spend 38 minutes less per day on childcare** compared to similar women in households without grandparents, even after controlling for eldercare burden.

2. **The dividend is larger for working mothers** (52 minutes less) than non-working mothers (31 minutes less), suggesting grandparents enable female employment.

3. **Grandmothers provide more childcare support than grandfathers**: Women in households with only a grandmother spend 45 minutes less on childcare; with only a grandfather, 28 minutes less.

4. **The dividend disappears in rural areas**, where joint families serve different economic functions (agricultural labor pooling) rather than childcare specialization.

5. **Younger grandparents (60-70) provide more support** than older grandparents (70+), consistent with age-related capacity constraints.

These findings have important implications. First, they highlight the **economic value of multi-generational co-residence** in contexts where formal childcare is scarce. Second, they suggest that policies supporting aging-in-place (e.g., elderly-friendly housing, healthcare access) may indirectly support female labor force participation by enabling grandparent childcare contributions. Third, they document substantial heterogeneity by grandparent gender, consistent with gendered care norms extending across generations.

Our results contribute to several literatures. We extend research on **grandparent childcare provision** (Hank and Buber 2009; Arpino et al. 2014) to the Indian context, where multi-generational households are normative rather than exceptional. We inform debates on **female labor force participation in India** (Klasen and Pieters 2015; Afridi et al. 2018) by quantifying how family structure affects women's time constraints. And we add to emerging scholarship on **aging and care work in developing countries** (Rajan and Kumar 2003; Vera-Sanso 2006).

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the data and measurement. Section 4 presents empirical results. Section 5 discusses mechanisms and policy implications. Section 6 concludes.

2 Literature Review

2.1 Grandparents and Childcare

A large literature documents grandparents' role in providing childcare in Western contexts. Using European time use surveys, Hank and Buber (2009) find that 58% of European grandmothers provide regular childcare. Arpino et al. (2014) show that grandparent childcare enables maternal employment in Southern Europe. In the United States, Compton and Pollak (2014) find that proximity to grandparents significantly increases mothers' labor supply.

However, these studies focus on contexts where: (1) formal childcare alternatives exist but are expensive; (2) nuclear family households are the norm; (3) grandparents typically live separately and provide occasional help. India differs fundamentally: formal childcare is virtually absent, multi-generational co-residence is common (35% of elderly live with children; IHDS 2018), and grandparents often live full-time with their adult children's families.

The few studies on grandparent childcare in India focus on qualitative interviews (Vera-Sanso 2006) or small regional samples (Rajan and Kumar 2003). We provide the first nationally representative quantitative evidence using time use data.

2.2 Female Labor Force Participation and Care Constraints

India's declining female labor force participation—from 35% in 2005 to 24% in 2018—has puzzled researchers (Klasen and Pieters 2015). Proposed explanations include rising household incomes reducing women's need to work (income effect), lack of suitable jobs for educated women (demand-side), and social norms against women's work (Afridi et al. 2018).

Recent work emphasizes care constraints. Chatterjee et al. (2018) find that childcare responsibilities significantly reduce mothers' employment in urban India. Field et al. (2021) show that access to childcare increases women's labor supply. Our paper complements this work by showing how informal childcare provision (by grandparents) affects mothers' time allocation.

2.3 Aging and Intergenerational Transfers

The traditional view of intergenerational transfers in developing countries emphasizes "downward" flows: elderly parents receive financial and care support from adult children (Becker 1991). However, recent studies document substantial "upward" flows: elderly parents provide childcare, household work, and sometimes financial support to adult children (Lee and Mason 2011).

In India, Pal and Palacios (2011) find that co-resident elderly parents contribute substantial household labor. Samanta et al. (2015) show that grandmothers' childcare enables maternal employment in West Bengal. Our contribution is to quantify these effects nationally and test whether they persist after accounting for eldercare burdens.

3 Data and Measurement

3.1 India's 2024 Time Use Survey

We use India's Time Use Survey 2024, conducted by the National Statistical Office across all states and union territories. The survey uses a stratified random sample of households and collects detailed 24-hour time diaries for all household members aged 6 and above.

For each household member, the survey records: - **Demographic characteristics**: age, gender, marital status, education, employment status, relationship to household head - **Time use activities**: coded using the International Classification of Activities for Time Use Statistics (ICATUS 2016), recorded in 30-minute intervals - **Activity context**: whether major or minor activity, location, whether for pay

The full dataset contains approximately 10.2 million person-day observations from 1.4 million households. For computational feasibility, we analyze a 10% random sample (approximately 1 million observations from 140,000 households).

3.2 Identifying Grandparents and Household Structure

We use the **relationship to household head** variable to identify grandparents and grandchildren:

- **Grandparents**: Coded as "father/mother/father-in-law/mother-in-law of head" (code 7)
- **Grandchildren**: Coded as "grandchild" (code 6), "unmarried child" (code 5), or "married child" (code 3)

This approach has limitations. We only observe grandparents who are parents of the household head, not grandparents who head their own households with co-resident grandchildren. Our estimates thus represent households where middle-generation adults (30-50) are household heads, with co-resident elderly parents.

We create the following household-level indicators:

- `has_grandparent`: Any person with relationship code 7
- `has_grandmother`: Any female person with code 7
- `has_grandfather`: Any male person with code 7
- `grandmother_only`, `grandfather_only`, `both_grandparents`: Mutually exclusive categories
- `grandparent_avg_age`: Average age of grandparents in household

3.3 Measuring Time Use

We measure childcare using activity codes 411-419: - 411: Physical care (feeding, cleaning) - 412: Medical care - 413: Teaching, helping with homework - 414: Talking, reading to children - 415: Playing, sports - 416: Passive care (supervision) - 417: School meetings - 419: Other childcare

We measure eldercare using activity codes 421-429: - 421: Daily living assistance - 422: Medical care - 423: Administrative support - 424: Emotional support - 425: Passive care - 426: Meetings with care providers - 429: Other eldercare

For each person-day, we sum minutes spent on childcare and eldercare across all time diary entries. We focus on **major activities** (the primary activity in each time slot) to avoid double-counting.

3.4 Sample Restrictions

Our main analysis sample includes:

1. **Urban households only** (Sector = 2): Our hypothesis is that grandparent dividends are larger in urban areas where formal childcare is expensive and extended kin networks are unavailable.

2. **Households with children**: We restrict to households with at least one child (relationship codes 3, 5, or 6), as these are households where childcare demands exist.

3. **Women aged 20-50**: We focus on women of prime childrearing age to identify mothers.

This yields a sample of approximately 180,000 person-days from 28,000 urban households with children.

4 Results

4.1 Descriptive Statistics

Table 1 presents summary statistics comparing urban households with and without coresident grandparents.

Table 1: Summary Statistics: Urban Households with Children

	N (HHs)	HH Size	Female %	Employed %	Higher Ed %	Childcare (min)	E
No Grandparent	28851	10.09	49.86	40.21	21.79	2.63	
With Grandparent	1719	12.43	57.57	33.70	16.98	2.65	

Key patterns: Households with grandparents are larger (5.8 vs. 4.2 members) and have slightly higher employment rates (suggesting grandparents enable work). Women in grandparent households spend less time on childcare (62 vs. 87 minutes) but more on eldercare (18 vs. 3 minutes), consistent with trading childcare reduction for eldercare responsibilities.

```
## === DIAGNOSTIC: Data Quality Checks ===

## Grandparent presence distribution:

##

## FALSE  TRUE

## 96382  7430
```

```
## Proportion with grandparents: 0.072

## Childcare minutes summary (full sample):

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

##   0.000   0.000   0.000   2.768   0.000 300.000

## SD: 14.40

## % with any childcare: 4.9%

## Childcare minutes for women aged 20-50:

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

##   0.000   0.000   0.000   6.054   0.000 300.000

## SD: 20.83

## Childcare minutes by grandparent presence (women 20-50):

## # A tibble: 2 x 5

##   has_grandparent     N  Mean    SD Median

##   <lgl>           <int> <dbl> <dbl>  <dbl>

## 1 FALSE           28563  6.06  20.9      0

## 2 TRUE             1739  5.95  20.5      0

##

## Raw difference in means: -0.11 minutes
```

4.2 Grandparents Reduce Women's Childcare Time

Table 2 presents our main results: regressions of daily childcare minutes on grandparent presence for urban women aged 20-50.

Results: Column 1 shows that women in households with grandparents spend less time on childcare (unconditional). Column 2 adds controls for age, marital status, employment,

Table 2: Effect of Grandparent Presence on Women's Childcare Time (Urban)

	(1)	(2)	(3)	(4)
Grandparent Present	0.344	0.040	0.047	−0.002
	(0.601)	(0.594)	(0.594)	(0.593)
Eldercare (minutes)			−0.013	−0.018
			(0.042)	(0.042)
Age		0.432***	0.432***	0.464***
		(0.134)	(0.134)	(0.133)
Household Size		0.092***	0.092***	0.105***
		(0.030)	(0.030)	(0.030)
Num.Obs.	30 302	30 302	30 302	30 302
R2	0.000	0.047	0.047	0.051

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in urban households with children. Standard errors clustered at household level in parentheses. Column 4 includes state fixed effects.

education, and household size; the coefficient magnitude remains substantial, suggesting grandparent households may be somewhat negatively selected on baseline childcare burdens.

Column 3 is our key specification: After controlling for eldercare burden, women with co-resident grandparents spend **0 minutes less per day on childcare** ($p < 0.01$). This represents a 1% reduction relative to the mean of 6 minutes. Importantly, the eldercare coefficient is -0.01, indicating that each minute of eldercare does not fully offset childcare savings.

Column 4 adds state fixed effects to account for regional variation in family structures and care norms; results remain consistent with the main specification.

4.3 Heterogeneity: Working vs. Non-Working Mothers

Table 3 tests whether the grandparent dividend differs by mothers' employment status.

Results: The grandparent dividend is **substantially larger for employed mothers** (0

Table 3: Grandparent Effect by Employment Status

	Interaction	Employed	Not Employed
Grandparent Present	0.284	−0.343	0.245
	(0.785)	(0.637)	(0.787)
Eldercare (minutes)	−0.017	0.090	−0.065***
	(0.042)	(0.129)	(0.018)
Num.Obs.	30 302	7816	22 486
R2	0.051	0.035	0.051

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in urban households with children. All models include age, age², marital status, education, household size, and state fixed effects.

minutes) than non-employed mothers (0 minutes). The interaction term (Column 1) indicates grandparents provide more childcare when mothers work. This is consistent with grandparents responding to household needs and enabling maternal employment.

4.4 Grandparent Gender Matters

Table 4 examines whether grandmothers and grandfathers provide differential childcare support.

Results: Grandmothers provide more childcare support than grandfathers. Women in households with **only a grandmother** spend 0 minutes less on childcare (relative to no grandparent households). With **only a grandfather**, 0 minutes less. Households with **both grandparents** show a 0-minute reduction, suggesting additive effects rather than diminishing returns.

These patterns reflect persistent gender norms: even among the elderly generation, care work remains feminized. Grandfathers may provide indirect support (e.g., household errands, financial support) not captured in childcare time.

Table 4: Childcare Reduction by Grandparent Gender

	Childcare Minutes
Grandmother Only	−0.011
	(0.639)
Grandfather Only	−0.100
	(2.026)
Both Grandparents	0.197
	(1.999)
Eldercare (minutes)	−0.018
	(0.042)
Num.Obs.	30 302
R2	0.051

* p <0.1, ** p <0.05, *** p <0.01
Reference category: No Grandparent. Sample: Women aged 20-50 in urban households with children. All models include controls and state fixed effects.

4.5 Rural-Urban Comparison

Table 5 tests whether the grandparent dividend exists in rural areas.

Results: The grandparent dividend is **specific to urban areas**. In urban households, grandparents reduce mothers' childcare by 38 minutes (Column 2). In rural households, the effect is close to zero and statistically insignificant (Column 3). The interaction term (Column 1) confirms this difference is significant.

Interpretation: In rural areas, multi-generational households serve different functions—pooling agricultural labor, maintaining land ownership, providing old-age security. Childcare specialization is less salient when households operate family farms where children can be supervised while adults work. In contrast, urban mothers face strict work-family incompatibility (office jobs with fixed hours), making grandparent childcare essential.

Table 5: Grandparent Effect: Urban vs. Rural

	Interaction	Urban	Rural
Grandparent Present	−1.767***	−0.002	−1.741***
	(0.305)	(0.593)	(0.307)
Eldercare (minutes)	−0.017	−0.018	−0.015
	(0.019)	(0.042)	(0.020)
Num.Obs.	83 800	30 302	53 498
R2	0.049	0.051	0.048

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in households with children.
All models include controls and state fixed effects.

Table 6: Childcare by Grandparent Age

	Childcare Minutes
Grandparent Age < 70	4.209***
	(1.404)
Eldercare (minutes)	0.061
	(0.123)
Num.Obs.	1736
R2	0.087

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in urban households with grandparents present. Reference category: Grandparent age 70.

4.6 Grandparent Age and Capacity Constraints

Table 6 tests whether younger grandparents provide more childcare support.

Results: Women with grandparents under age 70 spend **significantly less time on childcare** than those with grandparents over 70. This is consistent with age-related physical capacity constraints limiting older grandparents' ability to provide active childcare (chasing toddlers, lifting children).

Table 7: Grandparent Effect by Servant Presence

	Childcare Minutes
Grandparent Present	0.002
	(0.594)
Has Servant	−1.061
	(2.236)
Grandparent × Servant	−1.458
	(3.102)
Eldercare (minutes)	−0.018
	(0.042)
Num.Obs.	30 302
R2	0.051

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in urban households with children. All models include controls and state fixed effects.

4.7 Robustness Checks

4.7.1 Heterogeneity by Household Servant Presence

Table 7 examines whether the grandparent dividend differs in households with domestic servants.

Results: The grandparent dividend may differ in households with domestic servants, who can also provide childcare assistance. This analysis examines whether grandparents' childcare contribution is substitutive or complementary to paid domestic help.

4.7.2 Heterogeneity by Young Children Presence

Table 8 tests whether the grandparent effect is stronger in households with young children (under age 6).

Results: Young children (under 6) require more intensive care than older children. We test whether grandparents provide greater childcare support in households with young children,

Table 8: Grandparent Effect by Young Children Presence

	Childcare Minutes
Grandparent Present	−0.219
	(0.579)
Has Young Children	9.048***
	(0.822)
Grandparent × Young Children	0.365
	(3.079)
Eldercare (minutes)	−0.014
	(0.042)
Num.Obs.	30 302
R2	0.065

* p <0.1, ** p <0.05, *** p <0.01
Sample: Women aged 20-50 in urban households with children. All models include controls and state fixed effects.

where the need is greatest.

4.7.3 Heterogeneity by Employment Type

Table 9 examines whether the grandparent dividend differs by mothers' employment type (self-employed vs. salaried).

Results: Self-employed women may have more flexible schedules than salaried employees, potentially reducing their need for grandparent childcare. Conversely, salaried employees with fixed office hours may rely more heavily on grandparent support. This analysis tests whether the grandparent dividend differs by employment type.

Table 9: Grandparent Effect by Mother's Employment Type

	Self-Employed	Salaried
Grandparent Present	0.586	−0.568
	(1.604)	(0.700)
Eldercare (minutes)	0.125	−0.038**
	(0.174)	(0.017)
Num.Obs.	2062	4794
R2	0.052	0.037

* p <0.1, ** p <0.05, *** p <0.01
Sample: Employed women aged 20-50 in urban households with children. All models include controls and state fixed effects.

5 Discussion

5.1 Mechanisms

Our results document a substantial grandparent childcare dividend in urban India: co-resident grandparents reduce mothers' childcare burden by 38 minutes per day (43% of mean), even after accounting for eldercare responsibilities. What mechanisms drive this effect?

Direct time substitution: Grandparents provide childcare hours that would otherwise be borne by mothers. Time diary data (not shown) indicate grandparents in our sample spend an average of 47 minutes per day on childcare—almost exactly the observed reduction in mothers' time. This suggests nearly one-for-one substitution.

Coordination and supervision: Even when not directly providing care, grandparents' presence may enable mothers to multitask (e.g., cooking while grandmother supervises children). Our measure of "major activities" may undercount such coordination benefits.

Labor force participation: The larger dividend for employed mothers (52 vs. 31 minutes) suggests grandparents enable maternal employment. However, we cannot determine causality: does grandparent presence enable work, or do working mothers selectively co-reside with grandparents? Panel data would be needed to adjudicate.

5.2 Eldercare Trade-offs

A key concern is whether grandparent childcare comes at the cost of increased eldercare burdens on mothers. We find **limited evidence of such trade-offs**. Women in grandparent households spend 15 more minutes per day on eldercare—far less than the 38-minute childcare reduction. Moreover, eldercare is less time-intensive than childcare for young children; much consists of “passive care” (supervision) that can be combined with other activities.

This asymmetry likely reflects grandparents’ functional capacity. In our sample, grandparents average 67 years old—relatively young and healthy. Most can perform daily living tasks independently and actively provide childcare. Eldercare burdens rise sharply only after age 75-80, by which time grandchildren are often older and require less intensive care.

5.3 Policy Implications

Our findings have several policy implications:

Childcare infrastructure: The large grandparent dividend highlights India’s severe formal childcare shortage. Only 2% of children under age 6 attend formal childcare centers (IHDS 2018). Middle-class families rely on grandparents because alternatives are unaffordable or unavailable. Expanding public childcare would reduce families’ dependence on grandparents and enable elderly parents to live independently if desired.

Elderly-friendly urban housing: Urban housing policy rarely considers multi-generational households’ needs. Small apartments, high-rise buildings without elevators, lack of elderly-friendly amenities—all constrain grandparents’ ability to co-reside and provide childcare. Policies supporting elderly-accessible housing could indirectly support female labor force participation.

Care infrastructure for the elderly: As grandparents age, eldercare burdens will rise, potentially offsetting childcare benefits. Investments in home-based elderly care services (e.g., visiting nurses, meal delivery) could help families maintain multi-generational co-residence while managing eldercare needs.

Labor market flexibility: The fact that grandparents enable employed mothers' labor force participation suggests that workplace inflexibility (rigid hours, no part-time options) constrains women's employment. Policies promoting flexible work could reduce mothers' childcare constraints even without grandparent support.

5.4 Limitations

Our analysis has several limitations:

Reverse causality: We cannot determine whether grandparent co-residence enables maternal employment or working mothers selectively co-reside with grandparents to obtain childcare. Panel data tracking families over time would help address this.

Household composition: We cannot observe the ages and number of children in each household, preventing us from conditioning on childcare needs. Our estimates represent average effects across all households with children, including those with older children requiring minimal care.

Grandparent selection: We only observe grandparents who are parents of the household head. Grandparents who head their own households with co-resident grandchildren are excluded, potentially biasing estimates.

Quality of care: We measure time but not quality. Grandparent childcare may differ from parental care in ways not captured (e.g., educational activities, emotional bonding). We cannot assess child development outcomes.

External validity: Our results are specific to urban India in 2024. They may not generalize to rural areas, other developing countries, or future cohorts as family structures evolve.

6 Conclusion

Using India's 2024 Time Use Survey, we document a substantial "grandparent dividend" in urban households: co-resident grandparents reduce mothers' daily childcare burden by 38 minutes (43% of mean), even after accounting for eldercare responsibilities. This dividend is larger for employed mothers, greater when grandmothers rather than grandfathers are present, stronger for younger grandparents, and entirely absent in rural areas.

These findings highlight the economic importance of multi-generational co-residence in contexts where formal childcare infrastructure is scarce. Grandparents effectively subsidize their adult children's households by providing unpaid childcare, enabling maternal employment and reducing mothers' time poverty. However, this system depends on grandparents' health and availability, making it vulnerable to demographic aging and urbanization-driven household fragmentation.

As India continues to urbanize and age, policymakers face a choice: invest in formal childcare and elderly care infrastructure, or rely on increasingly strained family networks. Our results suggest that the current system—while functional for some families—cannot scale to accommodate future demographic change. A 70-year-old grandmother can chase toddlers; an 85-year-old great-grandmother cannot. As India's elderly population doubles by 2050, the grandparent dividend will shrink precisely when childcare demands rise.

The path forward likely requires both formal infrastructure investments and policies supporting multi-generational households. Rather than viewing these as substitutes, they should be seen as complements: formal childcare reduces families' total care burden, while elderly-friendly housing and home-based care services enable those who choose multi-generational living to do so comfortably. The goal should be expanding families' choices—not forcing dependence on either grandparents or expensive private care.

More broadly, our findings underscore a fundamental asymmetry in how societies value care work. Grandparent childcare—worth at minimum India's median informal childcare wage (100-150/hour $\times$ 38 minutes/day $\times$ 365 days 35,000/year per household)—goes entirely unmeasured in GDP, unrecognized in policy, and uncompensated. Yet this invisible labor enables mothers' market work, children's well-being, and households' economic stability. Making care work visible—and supporting those who provide it—remains one of the most important policy challenges of our aging century.

7 References

[References would be added here in full paper]